PHYS 253: Thermal Physics

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Abstract

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1 Introduction

1.1 Course Information

The textbook is *An Introduction to Thermal Physics* by Daniel V. Schroeder. Grading is divided as follows:

- 10% problem sets (~ 6, submitted as handwritten PDF scans)[1]
- 50% two midterms
- 40% final exam

[1] Only one or two problems on each assignment are marked for submission and grading.

2 Temperature

2.1 Basic Concepts

2.1.1 Systems and Boundaries

2.2 Equilibrium State

2.2.1 Equilibrium States and State Functions

2.2.2 Local Thermodynamic Equilibrium

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3 Reversible Processes and Work

3.1 Reversible Processes

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3.1.2 Reversible and Irreversible Expansion

3.1.3 Exact and Inexact Differentials

3.2 Work

3.2.1 Compression Work

3.3 Other Kinds of Work

4 The First Law of Thermodynamics

4.1 The Work of Thompson and Joule

Joseph Black, professor of medicine at the University of Glasgow, discovered the following principle:

Principle 1 (Black's Principle of Calorimetry)**.** *When two bodies are put in thermal contact, the heat lost by one is equal to the heat gained by the other.*

This suggested that heat was a conserved substance, called caloric.

Definition 1 (Caloric)**.** *Caloric is a hypothetical weightless, invisible fluid, once believed to be heat itself, conserved and flowing from hotter to colder bodies.*

Then came Benjamin Thompson, working in the arsenal in Munich, who noticed while boring cannons that friction produced seemingly unlimited heat:

Principle 2 (Thompson's Observation)**.** *Heat is not a finite substance such as caloric, and there is a direct relation between the work done and the heat produced.*

About 50 years later, James Joule finished the job with a careful series of experiments, using a tub containing a paddle wheel rotated by weights falling outside the tub. The water inside was stirred (irreversibly, because of turbulence), raising its temperature between two equilibrium states. The walls of the tub were insulating, so the work was performed under adiabatic conditions.

Definition 2 (Adiabatic Work)**.** *Adiabatic work is work done on a system enclosed by adiabatic (thermally insulating) walls, so that no heat is exchanged with the surroundings.*

Joule found:

Result 1 (Mechanical Equivalent of Heat)**.** 4.2 *kJ of work is required to raise the temperature of* 1 *kg of water by* 1 *K.*

Result 2 (Adiabatic Work)**.** *No matter how the adiabatic work is performed, the same amount of work is required to take a system between the same two equilibrium states.*

This second result is exactly the First Law of Thermodynamics, stated in the next subsection.

4.2 The First Law of Thermodynamics

Law 1 (First Law of Thermodynamics)**.** *If a thermally isolated system is brought from one equilibrium state to another, the work necessary to achieve this change is independent of the process used.*

This adiabatic work therefore defines a state function, the internal energy, through $\Delta U = W_{\text{adiabatic}}$. For a non-adiabatic process, the difference defines the heat:

$$\Delta U = W + Q, \tag{1}$$

where W is the work done on the system and Q the heat added to it.

Definition 3 (Heat)**.** *Heat Q is the nonmechanical exchange of energy between the system and the surroundings due to their temperature difference, flowing spontaneously from the hotter to the colder one.*

Remark 1 (Sign convention for Heat)**.** *Heat Q is positive when it enters the system and is negative when it leaves the system.*

Remark 2 (Differential Form)**.** *For an infinitesimal process, the first law takes the form*

$$dU = \delta W + \delta Q. \tag{2}$$

The δ are used to indicate that their integrals are in general path dependent (inexact differentials).

Definition 4 (Inexact Differential)**.** *An inexact differential δu is one for which no function f exists such that $\delta u = df$.*[2]

Proposition 1 (Work is an Inexact Differential)**.** *$\delta W = -P\,dV$ is inexact: its integral between two states depends on the path.*

Proof. Take two paths between the same states (P_1, V_1) and (P_2, V_2), with $P_1 \neq P_2$. Along path A, expand at P_1, then change pressure at constant V_2 (no work):

$$W_A = -P_1(V_2 - V_1).$$

Along path B, change pressure at constant V_1 (no work), then expand at P_2:

$$W_B = -P_2(V_2 - V_1).$$

Same end states, but $W_A \neq W_B$. In general, for an ideal gas write $\delta W = M\,dV + N\,dT$ with $M = -\frac{nRT}{V}$ and $N = 0$. Exactness would require $\left(\frac{\partial M}{\partial T}\right)_V = \left(\frac{\partial N}{\partial V}\right)_T$, but

$$\left(\frac{\partial M}{\partial T}\right)_V = -\frac{nR}{V} \neq 0 = \left(\frac{\partial N}{\partial V}\right)_T,$$

so no function $W(V, T)$ satisfies $dW = \delta W$. □

[2] Exactness, including the test $\left(\frac{\partial M}{\partial y}\right)_x = \left(\frac{\partial N}{\partial x}\right)_y$, is treated in detail in my Math 325 notes, in the section on exact equations.

4.3 Heat Capacity

4.3.1 C_V and C_P

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4.4.5 Heat Capacities from Equipartition

4.5 Ideal Gases and the First Law

Definition 5 (Free Expansion)**.** *Free expansion is the expansion of a gas into a vacuum, with no external pressure applied.*

- *The process is irreversible;*
- *There is no temperature change;*
- *No heat enters the system, since the walls are adiabatic;*
- *No work is done, since there is no external pressure;*

4.5.1 Internal Energy of an Ideal Gas
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6 Entropy

As previously mentioned, Sadi Carnot was the first to recognize that heat flowing from hot to cold can be used to produce work, but never all of it. The explanation behind this limit is entropy. The second law of thermodynamics can be restated in terms of entropy:

Law 2 (The Second Law of Thermodynamics (Entropy Formulation))**.** *the entropy of an isolated system never decreases with time. As a result, isolated systems evolve toward a thermodynamic equilibrium state where the entropy is maximized.*

6.1 The Clausius Inequality

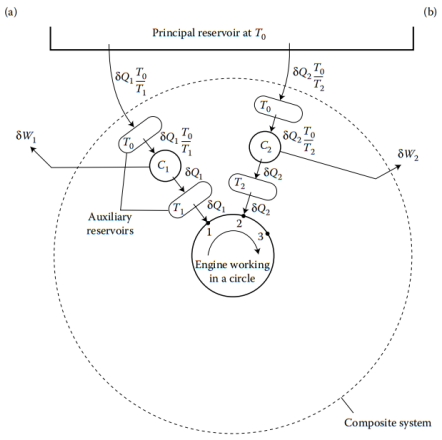


Figure 1: Clausius' thought experiment: Carnot engines route all heat exchanged by the gas through a single reservoir at T_0.

Could an engine run on heat drawn from the surrounding air alone, without burning fuel?

Remark 3 (Clausius' Thought Experiment)**.** *Consider Figure 1. Heat $\delta Q_1 \frac{T_0}{T_1}$ flows from the principal reservoir (PR) at T_0 (e.g. the surrounding air) to an auxiliary reservoir at T_0. A Carnot engine takes this heat, transforms part of it into work, and transfers the rest, δQ_1, to an auxiliary reservoir at the gas's temperature T_1, which passes it to the gas, changing its state from (P_1, V_1, T_1) to (P_2, V_2, T_2). Each auxiliary reservoir gives exactly what it receives, so it ends unchanged. This continues until the gas is back at its original state. The question is whether heat from a single reservoir can thus be fully converted into work, with no other effect on the surroundings.*

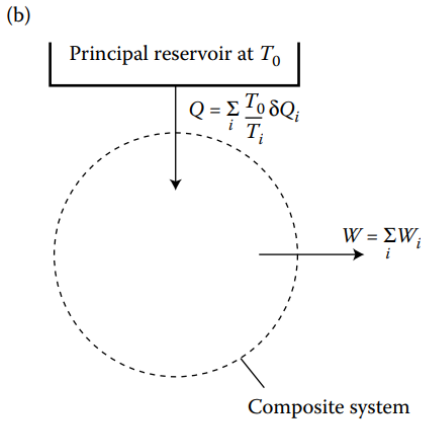


Figure 2: The composite system exchanges heat only with the reservoir at T_0.

After a full cycle, everything in the composite system (Figure 2) is back to its original state, so $\Delta U = 0$ and the first law gives

$$W = Q = T_0 \sum_i \frac{\delta Q_i}{T_i}, \tag{3}$$

where $W = \sum_i \delta W_i$ is the total work done and Q the heat drawn from T_0. A positive $W = Q$ is forbidden by:

Law 3 (Kelvin Statement of the Second Law of Thermodynamics)**.** *No process is possible whose sole result is the absorption of heat from a reservoir and the conversion of this heat into work.*

Hence $W = Q \leq 0$, and since $T_0 > 0$, taking the steps infinitesimal gives:

Law 4 (Clausius Inequality)**.** *For any cyclic process,*

$$\oint \frac{\delta Q}{T_{surr}} \leq 0, \tag{4}$$

where T_{surr} is the temperature of the auxiliary reservoirs supplying δQ_i (not of the system). If the cycle is reversible, $T_{surr} = T_{sys}$ and running it in reverse gives $\oint \frac{\delta Q}{T_{surr}} \geq 0$, so equality holds if and only if the cycle is reversible.

Remark 4. *Physically, an engine needs a hot source and a colder sink: a car takes heat from burning fuel, turns part of it into work, and dumps the rest into the colder air. With only one reservoir, there is nowhere to dump heat, so no net work is possible.*

6.2 Entropy

Suppose a system is taken along a reversible path Route_1 from an initial state i to a final state f and then back again to the initial state along another reversible path Route_2, completing a reversible cycle. Since the cycle is reversible, the Clausius inequality holds with equality:

$$\oint \frac{\delta Q_{\text{Route}}}{T} = \int_{i,\ \text{Route}_1}^{f} \frac{\delta Q_{\text{Route}}}{T} + \int_{f,\ \text{Route}_2}^{i} \frac{\delta Q_{\text{Route}}}{T} = 0,$$

therefore

$$\int_{i,\ \text{Route}_1}^{f} \frac{\delta Q_{\text{Route}}}{T} = -\int_{f,\ \text{Route}_2}^{i} \frac{\delta Q_{\text{Route}}}{T}.$$

Since Route_2 is reversible, traversing it from i to f instead flips the sign of every δQ, so $\int_{f,\ \text{Route}_2}^{i} = -\int_{i,\ \text{Route}_2}^{f}$ and

$$\int_{i,\ \text{Route}_1}^{f} \frac{\delta Q_{\text{Route}}}{T} = \int_{i,\ \text{Route}_2}^{f} \frac{\delta Q_{\text{Route}}}{T}.$$

The integral is the same along any reversible path between i and f, so it is path independent (depends only on the end states) and defines a state function.

Definition 6 (Entropy)**.** *The entropy S is the state function defined by*

$$\Delta S = S_f - S_i = \int_i^f \frac{\delta Q_{rev}}{T}, \tag{5}$$

where the integral is taken along any reversible path[3]*, for which $T_{sys} = T_{surr}$. Only entropy differences (ΔS) are defined, in units of J/K, using absolute temperature T in Kelvin. For an infinitesimal reversible process*

$$dS = \frac{\delta Q_{rev}}{T}. \tag{6}$$

Remark 5 (Entropy Change in Irreversible Processes)**.** *Since S is a state function, ΔS for an irreversible process equals ΔS along any reversible process between the same end states.*

Example (Entropy Change of Heating Water)**.** *Consider heating water from T_i to T_f, isobarically, through a series of reservoirs, each at $T + dT$ when the water is at T. The heat entering the water reversibly at each step is*

$$\delta Q_{rev} = C_P\, dT,$$

[3]Irreversibility comes from the temperature gap between system and source. The source must be at $T + dT$ hotter than the system at every moment.

where C_P is the water's heat capacity at constant pressure. Hence, the entropy change of the water is

$$dS = \frac{\delta Q_{rev}}{T} = \frac{C_P\, dT}{T},$$

or, for constant C_P,

$$\Delta S = C_P \int_{T_i}^{T_f} \frac{dT}{T} = C_P \ln \frac{T_f}{T_i}.$$

Definition 7 (Isentropic)**.** *A process is isentropic if the entropy of the system stays constant, $dS = 0$ at every step.*

Remark 6 (Isentropic Conditions)**.** *A process that is both adiabatic ($\delta Q = 0$) and reversible is isentropic, since $dS = \frac{\delta Q_{rev}}{T} = 0$. An adiabatic but irreversible process increases the system's entropy.*

6.3 The Principle of Increasing Entropy

Proposition 2 (Clausius Inequality for a Process)**.** *For any process between equilibrium states i and f,*

$$\int_i^f \frac{\delta Q}{T_{surr}} \le \Delta S, \qquad \textit{or, infinitesimally,} \qquad \frac{\delta Q}{T_{surr}} \le dS,$$

where the equality holds if the process is reversible, and then $T = T_{surr}$.[4]

Proof. Consider a cycle consisting of an irreversible path i to f followed by a reversible path f to i. The Clausius inequality gives

$$\int_i^f \frac{\delta Q}{T_{\text{surr}}} + \int_{\substack{f\\R}}^{i} \frac{\delta Q}{T_{\text{surr}}} \le 0 \implies \int_i^f \frac{\delta Q}{T_{\text{surr}}} \le -\int_{\substack{f\\R}}^{i} \frac{\delta Q}{T_{\text{surr}}} = \int_{\substack{i\\R}}^{f} \frac{\delta Q}{T} = \Delta S,$$

where the last equality is the definition of entropy, since the path is reversible. □

For an isolated system $\delta Q = 0$, so $\Delta S \ge 0$:

Law 5 (Principle of Increasing Entropy)**.** *The entropy of a thermally isolated system increases in any irreversible process and is constant in a reversible process; it never decreases.*

Remark 7. *Applied to the universe (system plus surroundings), a process can occur spontaneously only if the total entropy increases or stays the same. The principle applies to net changes of the entire system: the entropy of a part of a composite system, or of a non-isolated system, can decrease.*

[4] Here δQ is the actual heat, not the heat of the imaginary reversible process used to calculate dS.

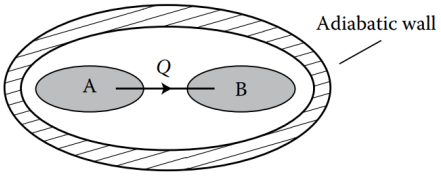


Figure 3: Although the entropy of a thermally isolated system can only increase or stay constant, the entropy of a part of the system can decrease.

Example. *Consider Figure 3, a system consisting of two subsystems which can exchange heat, the whole system being enclosed by adiabatic walls. If initially $T_A > T_B$, a small heat Q naturally flows from subsystem A to subsystem B. This reduces the entropy of subsystem A, $\Delta S_A = -\frac{Q}{T_A}$, while increasing the entropy of subsystem B, $\Delta S_B = \frac{Q}{T_B}$. But the total change is*

$$\Delta S = Q\left(\frac{1}{T_B} - \frac{1}{T_A}\right) > 0,$$

since $T_A > T_B$, so the entropy of the whole system increases.

Since the entropy of an isolated system can only increase, spontaneous changes continue until no further increase is possible, so equilibrium is reached when S is maximum.

Proposition 3 (Principle of Entropy Maximization)**.** *For a system that is thermally isolated from the surroundings, S tends to its maximum, whilst for a system that is totally isolated (thermally and mechanically), S tends to its maximum with U remaining constant.*

6.3.1 Entropy and the Arrow of Time

Remark 8 (Entropy and the Arrow of Time)**.** *Most physical laws are unchanged by replacing t with $-t$ (e.g. $F = m\frac{d^2\vec{r}}{dt^2}$), so they allow processes to run backward as well as forward. Yet a smashed teacup never reassembles itself: by the principle of increasing entropy (Law 5), the broken pieces have higher entropy than the unbroken cup. Entropy increase thus gives time its direction.*

6.4 Entropy–Temperature Diagrams

6.5 The Thermodynamic Identity

6.6 Other Examples of Entropy Calculations

6.6.1 Free Expansion

6.6.2 Measuring Entropies

7 Statistical Mechanics

7.1 Introduction to Probability and Statistics

Definition 8 (Microstate)**.** *A microstate is a specific configuration of the particles in a system.*

Definition 9 (Macrostate)**.** *A macrostate is a description of the system in terms of macroscopic properties, such as temperature, pressure, and volume. Different macrostates can be distinguished by a measurement, while different microstates of the same macrostate cannot.*

Assuming all microstates are equally probable, the probability of any macrostate is given by the multiplicity of that macrostate divided by the sum of all the multiplicities:

$$P(n) = \frac{\Omega(n)}{\sum_i \Omega(i)}. \tag{7}$$

For N two-state particles, the multiplicity $\Omega(N, n)$ of the macrostate with n out of the N particles in one of the two states can be calculated using the binomial coefficient:

$$\Omega(N, n) = \binom{N}{n} = \frac{N!}{n!(N-n)!}. \tag{8}$$

There are $N + 1$ macrostates and 2^N microstates in total:

$$\sum_{n=0}^{N} \binom{N}{n} = 2^N. \tag{9}$$

The distribution has mean $N/2$ and standard deviation $\sigma = \sqrt{N}/2$, so its relative width is

$$\frac{\sigma}{N} = \frac{1}{2\sqrt{N}}, \tag{10}$$

meaning the multiplicity becomes sharply peaked at $n = N/2$ as N grows.

For $N \gg 1$, factorials are evaluated with Stirling's approximation:

$$N! \approx N^N e^{-N} \sqrt{2\pi N}, \qquad \ln N! \approx N \ln N - N. \tag{11}$$

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8.7 Example Using a Maxwell Relation

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